# Rasheed Abdul Haq Kozhiparamban

rasheedabdulhaq@gmail.com
linkedin.com/in/rasheedabdulhaq
github.com/rasheedhaq
Bengaluru, Karnataka, India
+91 94004 72727

## Summary


- Senior AI/ML Engineer with a **Ph.D.** in Artificial Intelligence and 8+ years of experience delivering production AI systems across **Computer Vision**, **Industrial AI**, **Generative AI**, **Predictive Analytics**, and **Time-Series Forecasting**.
- Proven expertise in designing and deploying end-to-end AI solutions spanning data engineering, feature engineering, model development, validation, MLOps, inference, and human-in-the-loop workflows for enterprise and industrial applications.
- Hands-on experience with **Python**, **PyTorch**, **TensorFlow**, **OpenCV**, **YOLOv8**, **Azure ML**, **AWS Bedrock**, and modern LLM ecosystems for scalable AI development and deployment.
- Author of peer-reviewed IEEE publications and three granted Indian patents in Deep Learning, Virtual Sensing, and Predictive Analytics.


## Education

- **Ph.D.** in Artificial Intelligence (Deep Learning, Predictive Modeling & Time-Series Forecasting), National Institute of Technology (NIT), Puducherry, India, 2018 – 2024
- **M.Tech** in Electronics Design Technology, National Institute of Electronics & Information Technology (NIELIT), Calicut, India, 2012 – 2014
- **B.Tech** in Electronics and Communication Engineering, School of Engineering, CUSAT, Cochin, India, 2005 – 2009

## Experience

**Chevron Engine, Bengaluru, India: Senior Machine Learning Engineer**
**July 2025 – Present**

- Developed production-grade computer vision solutions for industrial inspection, asset integrity, and worker safety applications in the oil & gas sector.
- Advanced digital twin initiatives by integrating real-time sensor data with virtual asset models for condition monitoring, work planning, and faster troubleshooting.
- Evaluated and shortlisted computer vision vendors for EHS use cases through an RFP-style benchmark covering model accuracy (mAP/vmAP), latency, hazardous-area certifications (ATEX/IECEx), edge deployment, data privacy, and workflow integration.
- Led end-to-end AI solution development, including data ingestion, annotation strategy, model training, validation, deployment, inference, and human-in-the-loop review.
- Supported cloud-native MLOps practices including scalable model training, model registry, inference orchestration, data platform integration, and production monitoring.
- Improved model readiness through dataset audits, annotation quality reviews, class definition refinement, preprocessing recommendations, and validation gate design.
- Established annotation guidelines, quality review workflows, labeling standards, and SME validation processes to improve dataset consistency.
- Built high-resolution inspection imagery workflows involving image tiling, prediction stitching, confidence filtering, and structured output generation.
- Delivered safety-focused video analytics involving worker detection, activity recognition, PPE visibility, camera placement, occlusion handling, and data quality assessment.
- Authored technical documentation, architecture references, annotation handbooks, training material, and roadmap proposals while supporting knowledge transfer across teams.
- Collaborated with product owners, software engineers, data engineers, operations teams, inspectors, SMEs, and vendors to deliver deployable AI solutions and identify future computer vision opportunities.

**SONY, Bengaluru, India: Lead AI/ML Engineer**
**April 2025 – June 2025** (3-month contract engagement)

- Built secure **RAG pipelines** using AWS Bedrock and Claude with enterprise data from **SharePoint and Confluence** via MCP, enforcing citation tracking and zero internet leakage.
- Integrated and tuned **Jira/Confluence MCP clients**, using AWS Secrets Manager for tokenized access and validated `--space/project` parameters.
- Developed **LLM tools for task analysis** and automated risk scanners, streamlining deployments across Windows and AWS EC2 environments.

**DIMAAG-AI, Bengaluru, Karnataka, India: Senior Software Engineer AI/ML**
**February 2023 – February 2025** (2 years 1 month)

- Developed production-ready computer vision solutions for industrial and agricultural applications, leveraging **YOLOv8** for object detection and vision-LLM pipelines for advanced feature extraction.
- Deployed deep learning techniques for real-time **action recognition** and video analytics, implementing models such as **SlowFast**.
- Created methods for **Field of View (FOV)** validation to maintain camera alignment, combining feature matching, optical flow, and statistical comparisons.
- Built AI pipelines for **object segmentation**, classification, and feature extraction using segmentation models (**Unet**, **DeepLabV3**, **PSPNet**).
- Processed **3D point clouds** to generate segmentation masks, delivering accurate feature extraction for automation systems.
- Applied **Random Forest**, **Gradient Boosting**, and **MLP** classifiers for robust, high-accuracy predictions across diverse datasets.
- Enhanced data quality and balance via **image clustering** and model evaluation techniques, followed by **data augmentation** and hyperparameter tuning to improve model performance.
- Integrated **statistical analysis** and geometric transformations to optimize algorithmic precision for production environments.
- Implemented **LLM-based retrieval systems** to extract structured information from PDFs using **Vector Databases** (FAISS, Pinecone) and **Retrieval-Augmented Generation (RAG)**.
- Orchestrated **agentic workflows** via **LangChain** for dynamic query generation, structured output validation, and multi-step reasoning with LLMs.
- Developed prototypes leveraging **Generative AI** for summarizing SOP documents and automating question-answering systems, ensuring secure role-based access to sensitive data.
- Utilized **Azure AI services** (Azure Machine Learning, Cognitive Services) to streamline model training, deployment, and monitoring.
- Deployed **MLOps** workflows using **MLflow** for experiment tracking, model versioning, and containerized deployment on **Azure**.
- Drove end-to-end design, execution, and evaluation of AI model experiments, collaborating with cross-functional teams and mentoring junior engineers on deep learning best practices.

**National Institute of Technology, Puducherry, India: Senior Research Fellow**
**July 2018 – January 2023** (4 years 7 months)

- Developed virtual sensing and predictive analytics solutions for multivariate time-series forecasting of water quality parameters using deep learning.
- Engineered and optimized deep learning models (LSTM, GRU, Bi-LSTM) for multivariate time-series forecasting, improving prediction accuracy for aquaculture management.
- Proposed and implemented hybrid CNN-LSTM and CNN-GRU architectures for real-time predictive modeling, balancing forecasting accuracy and computational efficiency.
- Researched hyperparameter optimization (learning rate, epochs, batch size) to maximize model performance, achieving significant reductions in computation time.
- Applied **attention mechanisms** to deep learning models to enhance the accuracy of water quality predictions by focusing on the most relevant features in time-series data.

- Published peer-reviewed journal papers on predictive modeling, virtual sensing, and deep learning for time-series forecasting.
- Filed and was granted **patents** for new methods in virtual sensing and hybrid deep learning systems for aquaculture water quality monitoring.
- Conducted real-world experiments on aquaculture datasets (e.g., ADAK and MAC datasets) to validate the accuracy of the developed models.
- Designed and implemented an IoT-based low-cost water quality monitoring system, using sensors and cloud-based virtual sensing to provide real-time data for farmers.
- Built and validated water quality estimation models based on low-cost sensor data, reducing dependency on expensive sensors while maintaining high accuracy in dissolved oxygen prediction.

**Patents:**

1. Rasheed Abdul Haq K. P. and Harigovindan V. P. , "VIRTUAL SENSING SYSTEM FOR WATER QUALITY MONITORING AND METHOD THEREOF", Indian Patent No.: 397595, Date of Grant: May 25, 2022.
2. Rasheed Abdul Haq K. P. and Harigovindan V. P. , "DEEP LEARNING BASED VIRTUAL SENSING METHOD AND SYSTEM FOR LOW-COST AQUACULTURE WATER QUALITY MONITORING", Indian Patent No.: 552671, Date of Grant: October 21, 2024.
3. Rasheed Abdul Haq K. P. and Harigovindan V. P. , "HYBRID DEEP LEARNING BASED METHOD AND SYSTEM FOR AQUACULTURE WATER QUALITY PREDICTION", Indian Patent No.: 579506, Date of Grant: January 30, 2026.

**Journals:**

1. Rasheed Abdul Haq K. P. and Harigovindan V. P. "Deep Learning Enhanced Virtual Sensing: A Cost-Effective Approach for Water Quality Monitoring in Aquaculture" *IEEE Transactions on AgriFood Electronics*, June 2025.
2. Rasheed Abdul Haq K. P. and Harigovindan V. P., "Water Quality Prediction for Smart Aquaculture Using Hybrid Deep Learning Models", *IEEE Access*, vol. 10, pp. 60078-60098, June 2022.
3. Rasheed Abdul Haq K. P. and Harigovindan V. P., "Water Quality Prediction System based on Adam Optimised LSTM Neural Network for Aquaculture: A Case Study in Kerala, India", *J. Inst. Eng. India Ser. B*, *Springer*, 103, 2177–2188, October 2022.
4. Rasheed Abdul Haq K. P. and Harigovindan V. P., "Accurate Dissolved Oxygen Prediction for Aquaculture Using Stacked Ensemble Machine Learning Model", *Natl. Acad. Sci. Lett.*, *Springer*, 46, 203–207, January 2023.

**MEA Engineering College, Perinthalmanna, Kerala, India: Assistant Professor, Department of ECE**
**June 2015 – July 2018** (3 years 1 month)

- Delivered undergraduate and postgraduate courses in Digital Signal Processing, Signals & Systems, Digital System Design, Basics of Electronics Engineering, Electronic Circuits, and Communication Systems, while developing structured course materials aligned with the APJ Abdul Kalam Technological University (KTU) curriculum.
- Led the department's transition from the Calicut University syllabus to the KTU curriculum by preparing course files, lecture notes, assessment plans, question banks, laboratory manuals, and outcome-based evaluation documents.
- Served as Class Tutor for the 2014–2018 Electronics & Communication Engineering batch, managing student mentoring, attendance analytics, academic performance tracking, parent interactions, and intervention plans for academically at-risk students.
- Coordinated B.Tech Mini Projects, B.Tech Seminars, and M.Tech Seminar activities by organizing student groups, assigning faculty mentors, scheduling evaluations, and consolidating academic assessments.
- Guided undergraduate engineering projects in embedded systems and wireless technologies while mentoring students in technical writing, project execution, and research presentation.
- Conducted laboratory sessions in Basic Electronics, Linear Integrated Circuits, and Communication Systems, and served as an External Laboratory Examiner for Electronics laboratories under the university examination system.
- Developed spreadsheet-based academic tracking systems for attendance, internal assessment, and departmental reporting, improving the efficiency and accuracy of academic administration.

**Way in Electronic Solutions, Calicut, India: Founder & Senior Embedded Software Engineer**
**August 2014 – June 2017 (Part-time while serving as Assistant Professor)** (3 years 10 months)

- Co-founded an embedded systems startup delivering custom industrial automation, smart parking, medical-device testing, and intelligent monitoring solutions from concept through deployment.
- Led end-to-end product development, including requirements analysis, system architecture, embedded firmware development, hardware integration, testing, technical documentation, deployment, and customer support.
- Designed and developed embedded products including the Intelligent Parking Guidance System (IPGS), Ultrasound Probe Test System, GSM-based Security System, Digital Signage solutions, and Animal Health Monitoring System (AHMS).
- Engineered Embedded C firmware for 8051-based systems and Raspberry Pi applications, integrating RS-485/UART communication, ultrasonic sensors, LED driver interfaces, and Linux-based embedded platforms.
- Built web-based monitoring and management applications using PHP, MySQL, JavaScript, HTML/CSS, and XAMPP for real-time visualization, remote monitoring, and data management.
- Authored high-level design documents (HLDs), coding standards, technical proposals, quotations, business plans, and customer presentations while supporting product planning and engineering best practices.
- Collaborated directly with customers, technology partners, and multidisciplinary teams to translate business requirements into deployable engineering solutions while contributing to startup operations, recruitment, and business development.

**M. N. Dastur & Co. (P) Ltd., Bengaluru, India: DSP Intern (Embedded Image Processing)**
**September 2013 – July 2014** (11 months)

- Engineered an Automatic Number Plate Recognition (ANPR) system for Indian vehicles using Computer Vision and Digital Image Processing as part of an M.Tech internship project targeting embedded DSP/ARM platforms.
- Designed and implemented the complete recognition pipeline covering license plate localization, character segmentation, Optical Character Recognition (OCR), and Support Vector Machine (SVM)-based classification using MATLAB, OpenCV, and C++.
- Implemented image preprocessing, Sobel edge detection, Otsu thresholding, morphological operations, contour analysis, histogram equalization, affine rotation correction, and feature extraction to improve recognition robustness.
- Ported and evaluated the OpenCV implementation on Ubuntu Linux running on the ARM-based BeagleBoard-xM platform, addressing embedded deployment and cross-compilation challenges.
- Authored core C++ modules, MATLAB algorithms, training utilities, technical documentation, and project reports while validating the solution through iterative prototype development and engineering reviews.

**Bharti Airtel Ltd., Kerala, India: Network Surveillance Engineer**
**January 2010 – February 2012** (2 years 2 months)

- Monitored nationwide telecom network health, service availability, alarms, and key performance indicators (KPIs) using Operations & Maintenance Center (OMC-R) network management tools.
- Performed network fault detection, troubleshooting, root cause analysis, and performance optimization by coordinating with field engineers, vendors, and cross-functional operations teams to minimize service disruptions.
- Coordinated planned network outages, preventive maintenance activities, and vendor-recommended corrective actions while ensuring controlled execution with minimal customer impact.
- Provided remote technical support for fault investigation, managed alarm escalation processes, classified network outages, and ensured timely incident resolution in accordance with established escalation procedures.
- Analyzed network alarm trends and generated daily, weekly, and monthly operational reports to support performance monitoring, reliability improvements, and continuous network optimization.

## Pet AI Projects

**Arogyamitra — AI medical triage assistant for Telegram (Kerala, India):** **2025**
*LLM-driven chatbot that collects demographics and symptoms, triages risk, and recommends specialists.*

- Delivered production-ready Telegram bot with structured multi-field demographic and symptom collection.
- Implemented multi-provider LLM fallback and persistent rate-limit tracking for robust uptime.
- Built precise doctor-matching with exclude-keyword logic to prevent misclassification.

*Stack: Python, python-telegram-bot, Groq (LLM), OpenRouter/Gemini, HuggingFace, CSV (data), SQLite (planned), Railway (deployment) Links: GitHub | Demo*

## Skills

- **Programming Languages:** Python, C++, MATLAB
- **AI/ML Expertise:** Predictive Modeling, Time-Series Forecasting, Statistical Learning, Machine Learning, Deep Learning, Computer Vision, Digital Twins, NLP, Generative AI, LLMs, Retrieval-Augmented Generation (RAG), Agentic AI
- **Libraries/Frameworks:** PyTorch, TensorFlow, Keras, Scikit-Learn, YOLOv8, PyTorchVideo, Hugging Face Transformers, LangChain, NumPy, Pandas, Matplotlib, NLTK, spaCy
- **MLOps & Cloud:** Azure ML, Azure AI Services, AWS Bedrock, MLflow, Git
- **Vector Databases & Retrieval:** FAISS, Pinecone
- **Operating Systems:** Linux, Windows
- **Professional Skills:** Technical Leadership, Cross-functional Collaboration, Technical Mentoring, Stakeholder Communication, Technical Documentation

## References

Available upon request.